NCERT Mathematics — Class 6

Chapter 7: Fractions

7.1 What is a Fraction?

A fraction means a part of a whole or collection. A whole can be a single object or a group of objects. The whole must be divided into equal parts. In a fraction **a/b**, **a** is the **numerator** and **b** is the **denominator**.

Types of Fractions:
• **Proper Fraction:** Numerator is strictly less than denominator (e.g. 2/3, 5/8). Always < 1.
• **Improper Fraction:** Numerator is greater than or equal to denominator (e.g. 7/4, 9/5). Always ≥ 1.
• **Mixed Fraction:** A combination of a whole number and a proper fraction (e.g. 1 3/4).

7.2 Equivalent Fractions and Simplest Form

To find an equivalent fraction, we may multiply or divide both the numerator and the denominator of a given fraction by the same non-zero number.

Simplest Form: A fraction is said to be in simplest (or lowest) form if its numerator and denominator have no common factor except 1 (i.e. HCF is 1).
Example: 36/48 = (36 ÷ 12) / (48 ÷ 12) = 3/4.

7.3 Addition and Subtraction of Fractions

Like Fractions: Fractions with the same denominator. To add like fractions, add the numerators and keep the common denominator: **a/d + b/d = (a + b)/d**.
Unlike Fractions: Fractions with different denominators. Convert them to equivalent fractions with common denominator using LCM, then add or subtract.

Example: Sarita bought 2/5 metre of ribbon and Lalita 3/4 metre of ribbon. What was the total length of ribbon they bought?

Solution:
Length bought by Sarita = 2/5 m
Length bought by Lalita = 3/4 m
Total length = 2/5 + 3/4.
LCM of 5 and 4 is 20.
2/5 = (2 × 4)/20 = 8/20.
3/4 = (3 × 5)/20 = 15/20.
Total = 8/20 + 15/20 = 23/20 = 1 3/20 metres.